# Grade 10 – Term 1
# Decision-Making Under Uncertainty
# Exam Solutions – Version 2D

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## Learning objective

Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty.

## Criteria assessment

This exam assesses C8 and C9, each worth 6 points. A **minimum of 4 points per criterion** is required to pass. Show all relevant calculations and justify your answers. Partial credit may be awarded for correct procedures despite arithmetic errors; answers without demonstrated procedures receive no credit. A 1-point penalty per criterion applies if the student's name is missing or the assessment is submitted late. C10 is assessed separately.

| Question | C8 | C9 |
|---|---|---|
| 1 | 3 | – |
| 2 | 3 | – |
| 3 | – | 2 |
| 4 | – | 2 |
| 5 | – | 2 |
| **Total** | **6** | **6** |

## Evaluated criteria

**C8** Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations.

**C9** Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.

**C10** Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context.

## 1 Camera Rental Plans [C8-3]

A camera rental studio selects its inventory under uncertain demand. Profits are in thousand USD.

| Alternative | High | Normal | Low |
|---|---|---|---|
| $P(S_j)$ | 0.3 | 0.4 | 0.3 |
| Essential ($A_1$) | 95 | 88 | 62 |
| Shared ($A_2$) | 135 | 148 | 32 |
| Extensive ($A_3$) | 235 | 46 | -38 |
| Modular ($A_4$) | 165 | 125 | -18 |

Decision statistics (Regret = maximum regret):

| Alt. | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 95 | 62 | 140 | 82.3 |
| $A_2$ | 148 | 32 | 100 | 109.3 |
| $A_3$ | 235 | -38 | 102 | 77.5 |
| $A_4$ | 165 | -18 | 80 | 94.1 |

**(a)** Using the original camera rental payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

**(b)** A subsidy $s$ (integer, $0 \leq s \leq 90$) increases the low-demand payoffs of $A_3$ and $A_4$ by $s$. An alternative is eligible only if all its payoffs are nonnegative. Determine each camera rental plan's eligibility based on the value of $s$. **[0.5]**

**(c)** Among eligible camera rental plans, find the smallest allowed subsidy $s$ at which the Maximin recommendation changes from $A_1$ to $A_4$ alone. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $0 \leq s \leq 90$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

## (a) Initial recommendations [1]

| Criterion | Recommended alternative(s) | Deciding statistic and value |
|---|---|---|
| Maximax | $A_3$ | Row maximum: 235 |
| Maximin | $A_1$ | Row minimum: 62 |
| Minimax Regret | $A_4$ | Maximum regret: 80 |
| EMV/Bayes | $A_2$ | EMV: 109.3 |

All deciding values are in thousand USD. Maximax pursues the greatest upside; Maximin protects the worst outcome; Minimax Regret limits missed profit; EMV uses the forecast probabilities to maximize expected profit. Regret benchmarks use all four original alternatives, with state bests $(235, 148, 62)$; no eligibility restriction applies in (a).



## (b) Subsidy and eligibility [0.5]

The updated payoff table is:

| Alternative | High | Normal | Low |
|---|---|---|---|
| $P(S_j)$ | 0.3 | 0.4 | 0.3 |
| Essential ($A_1$) | 95 | 88 | 62 |
| Shared ($A_2$) | 135 | 148 | 32 |
| Extensive ($A_3$) | 235 | 46 | $-38+s$ |
| Modular ($A_4$) | 165 | 125 | $-18+s$ |

The updated decision statistics are:

| Alt. | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 95 | 62 | 140 | 82.3 |
| $A_2$ | 148 | 32 | 100 | 109.3 |
| $A_3$ | 235 | $\min(46, -38+s)$ | $\max(102, 100-s)$ | $77.5+0.3s$ |
| $A_4$ | 165 | $-18+s$ | $\max(70, 80-s)$ | $94.1+0.3s$ |

An alternative is eligible only if all its payoffs are nonnegative. For integer subsidies $0 \le s \le 90$:

| Alternative | Eligibility condition | Eligible subsidies |
|---|---|---|
| $A_1$ | $62 \ge 0$ | $0 \le s \le 90$ |
| $A_2$ | $32 \ge 0$ | $0 \le s \le 90$ |
| $A_3$ | $-38+s \ge 0$ | $38 \le s \le 90$ |
| $A_4$ | $-18+s \ge 0$ | $18 \le s \le 90$ |

Thus, the eligible sets are:

$$\begin{aligned} 0 \le s \le 17: &\quad \{A_1, A_2\}, \\ 18 \le s \le 37: &\quad \{A_4, A_1, A_2\}, \\ 38 \le s \le 90: &\quad \{A_3, A_4, A_1, A_2\}. \end{aligned}$$

### (c) Maximin change under subsidy $s$ [1.5]

The minimum payoffs are

$$m_1 = 62, \quad m_2 = 32, \quad m_3(s) = \min(46, -38 + s), \quad m_4(s) = -18 + s.$$

Since $m_3(s) \le 46 < 62 = m_1$ and $m_2 = 32 < 62$ throughout the allowed range, only $A_4$ can challenge $A_1$.

For $A_4$ to beat $A_1$:

$$-18 + s > 62 \quad \Longleftrightarrow \quad s > 80.$$

Thus, $A_1$ is the sole Maximin choice for $0 \le s \le 79$; at $s = 80$, $A_4$ and $A_1$ tie with minimum payoff 62; and $A_4$ is the sole Maximin choice for $81 \le s \le 90$. The tie at $s = 80$ first changes the set of recommended plans; a strict change to $A_4$ alone requires $s > 80$. The integer domain contains a smallest value satisfying this condition. Therefore, the smallest allowed integer subsidy giving a strict change is

$$\boxed{s = 81}.$$

At $s = 81$, all four alternatives are eligible and their minimum payoffs are $62, 32, 43, 63$, respectively. Hence $A_4$ wins uniquely with a guaranteed profit of 63 thousand USD.

## 2 Outdoor Cinema Refreshments [C8-3]

A cinema manager selects a refreshment kiosk under uncertain audience size. Profits are in hundred USD.

| Alternative | High | Low |
|---|---|---|
| $P(S_j)$ | 0.5 | 0.5 |
| Portable ($C_1$) | 92 | 60 |
| Full-service ($C_2$) | 150 | 16 |
| Deluxe ($C_3$) | 176 | -16 |
| Fixed ($C_4$) | 136 | 46 |

Decision statistics (Regret = maximum regret):

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $C_1$ | 92 | 60 | 84 | 76 |
| $C_2$ | 150 | 16 | 44 | 83 |
| $C_3$ | 176 | -16 | 76 | 80 |
| $C_4$ | 136 | 46 | 40 | 91 |

**(a)** Using the original refreshment kiosk payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

**(b)** A sponsor replaces $C_1$'s large-audience payoff of 92 with $g$ (integer, $92 \leq g \leq 156$). An alternative is eligible only if all its payoffs are at least 38. Determine each refreshment kiosk's eligibility based on the value of $g$. **[0.5]**

**(c)** Among eligible refreshment kiosks, find the smallest allowed sponsored payoff $g$ at which the Minimax Regret recommendation changes from $C_4$ to $C_1$ alone. Recalculate regret benchmarks using only eligible kiosks. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $92 \leq g \leq 156$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

## (a) Initial recommendations [1]

| Criterion | Recommended alternative(s) | Deciding statistic and value |
|---|---|---|
| Maximax | $C_3$ | Row maximum: 176 |
| Maximin | $C_1$ | Row minimum: 60 |
| Minimax Regret | $C_4$ | Maximum regret: 40 |
| EMV/Bayes | $C_4$ | EMV: 91 |

All values are in hundred USD. Regret benchmarks use all four original kiosks, with state bests $(176, 60)$; no eligibility restriction applies in (a). Only EMV uses the probabilities; the other criteria compare upside, downside, or missed profit.

## (b) Sponsored payoff and eligibility [0.5]

The updated payoff table is:

| Alternative | High | Low |
|---|---|---|
| $P(S_j)$ | 0.5 | 0.5 |
| Portable ($C_1$) | $g$ | 60 |
| Full-service ($C_2$) | 150 | 16 |
| Deluxe ($C_3$) | 176 | -16 |
| Fixed ($C_4$) | 136 | 46 |

The updated decision statistics, before applying eligibility restrictions, are:

| Alt. | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $C_1$ | $g$ | 60 | $176 - g$ | $0.5g + 30$ |
| $C_2$ | 150 | 16 | 44 | 83 |
| $C_3$ | 176 | -16 | 76 | 80 |
| $C_4$ | 136 | 46 | 40 | 91 |

An alternative is eligible only if both payoffs are at least 38. For integers $92 \leq g \leq 156$:

| Alternative | Eligibility condition | Eligible values |
|---|---|---|
| $C_1$ | $g \geq 38,\ 60 \geq 38$ | $92 \leq g \leq 156$ |
| $C_2$ | $16 < 38$ | None |
| $C_3$ | $-16 < 38$ | None |
| $C_4$ | $136 \geq 38,\ 46 \geq 38$ | $92 \leq g \leq 156$ |

The eligible set is always $\{C_1, C_4\}$.

### (c) Minimax Regret change under sponsored payoff $g$ [1.5]

Recalculate regrets using only these alternatives, with state benchmarks

$$b_H = \max\{g, 136\}, \qquad b_L = 60.$$

Their maximum regrets are

$$R(C_1) = \max\{136 - g, 0\},$$
$$R(C_4) = \max\{g - 136, 14\}.$$

For $92 \leq g \leq 136$, these become

$$R(C_1) = 136 - g, \qquad R(C_4) = 14.$$

For $C_1$ to be the sole Minimax Regret choice:

$$136 - g < 14 \quad \Longleftrightarrow \quad g > 122.$$

At $g = 122$, both alternatives have maximum regret 14. Thus, $C_4$ is the sole Minimax Regret choice for $92 \leq g \leq 121$; at $g = 122$, the recommendation becomes the tie $\{C_1, C_4\}$; and $C_1$ is the sole choice for $123 \leq g \leq 156$. For $g > 136$, $C_1$ has zero regret and $C_4$ has regret at least 14, so no further change occurs. The tie first changes the set of recommended kiosks, but a strict change to $C_1$ alone requires $g > 122$. The integer domain contains a smallest value satisfying this condition. Therefore, the smallest allowed integer giving a strict change is

$$g = 123\ .$$

At this value, the eligible payoff table is:

| Alternative | High | Low |
|---|---|---|
| $C_1$ | 123 | 60 |
| $C_4$ | 136 | 46 |

The corresponding maximum regrets are

$$R(C_1) = 13, \qquad R(C_4) = 14.$$

Thus, $C_1$ wins uniquely with $g = 123$, corresponding to \$12,300 in large-audience profit.

## 3 Garden Centre Stock [C9-2]

A garden centre selects a seedling stock plan $D_1$–$D_6$. Net profits are in hundred USD. The decision statistics are summarized below:

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $D_1$ | 120 | 56 | 48 | 102.4 |
| $D_2$ | 136 | 40 | 32 | 113.6 |
| $D_3$ | 104 | 72 | 64 | 92.8 |
| $D_4$ | 152 | 24 | 48 | 113.6 |
| $D_5$ | 88 | 56 | 80 | 76.8 |
| $D_6$ | 168 | 8 | 64 | 113.6 |

Apply these stages in order:

1. EMV/Bayes: Best 3.
2. Maximax: Best 2.
3. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

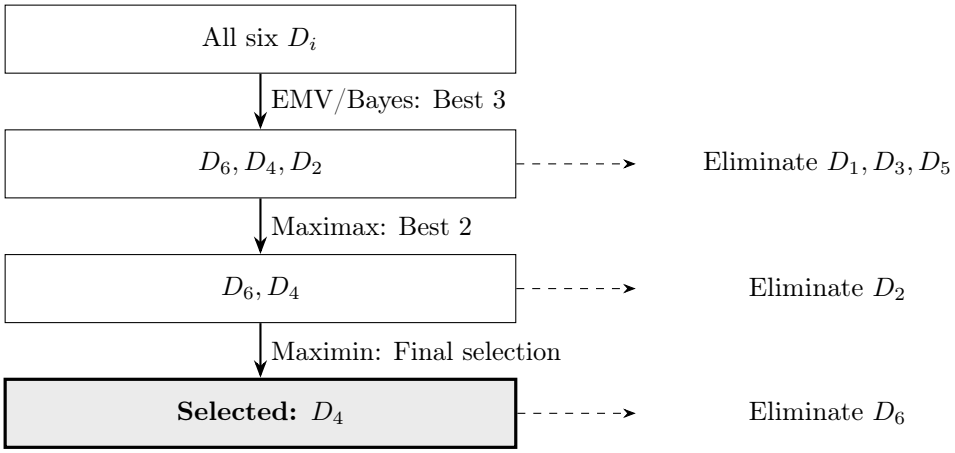
### Supporting filtering table

| Step | Rule | Survivors (relevant values) | Eliminated (values) |
|---|---|---|---|
| 0 | Initial alternatives | $D_6, D_4, D_2, D_1, D_3, D_5$ | — |
| 1 | EMV/Bayes: Best 3 | $D_6(113.6)$, $D_4(113.6)$, $D_2(113.6)$ | $D_1(102.4)$, $D_3(92.8)$, $D_5(76.8)$ |
| 2 | Maximax: Best 2 | $D_6(168)$, $D_4(152)$ | $D_2(136)$ |
| 3 | Maximin: Final selection | **Selected:** $D_4(24)$ | $D_6(8)$ |

The highest EMV, 113.6, is shared by $D_6, D_4, D_2$; all three fill the Best 3 places. Maximax retains $D_6, D_4$ with maxima 168 and 152, eliminating $D_2$ with 136. Maximin selects $D_4$ because $24 > 8$. There are no cutoff or final ties beyond the shared first-stage EMV.



**Filtering tree and interpretation**




The garden centre selects $D_4$: it survives the EMV and Maximax filters, then its minimum profit of 24 hundred USD exceeds $D_6$'s 8.

## 4 Climbing Gym Expansion [C9-2]

A climbing gym selects an expansion plan $F_1$–$F_6$. Net profits are in thousand USD. The decision statistics are summarized below:

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $F_1$ | 253 | 71 | 65 | 184.1 |
| $F_2$ | 201 | 123 | 117 | 172.4 |
| $F_3$ | 227 | 97 | 91 | 195.8 |
| $F_4$ | 318 | 6 | 117 | 164.6 |
| $F_5$ | 175 | 110 | 143 | 149 |
| $F_6$ | 279 | 45 | 78 | 173.7 |

Apply these stages in order:

1. Minimax Regret: Best 5.
2. EMV/Bayes: Best 4.
3. Maximax: Best 2.
4. Maximin: Final selection.

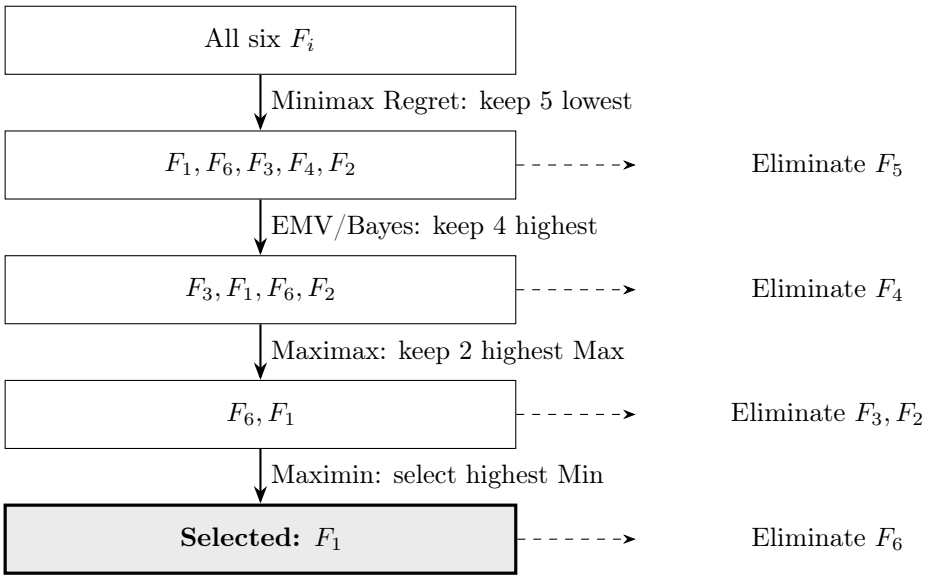
Illustrate each filtering stage, showing survivors and eliminations.

### Supporting filtering table

| Step | Rule | Survivors (relevant values) | Eliminated (values) |
|---|---|---|---|
| 0 | Initial alternatives | $F_4, F_6, F_1, F_3, F_2, F_5$ | — |
| 1 | Minimax Regret: keep 5 lowest | $F_1(65)$, $F_6(78)$, $F_3(91)$, $F_4(117)$, $F_2(117)$ | $F_5(143)$ |
| 2 | EMV/Bayes: keep 4 highest | $F_3(195.8)$, $F_1(184.1)$, $F_6(173.7)$, $F_2(172.4)$ | $F_4(164.6)$ |
| 3 | Maximax: keep 2 highest Max | $F_6(279)$, $F_1(253)$ | $F_3(227)$, $F_2(201)$ |
| 4 | Maximin: select highest Min | **Selected:** $F_1(71)$ | $F_6(45)$ |

At the first cutoff, $F_4$ and $F_2$ both have regret 117, and both survive. Later cutoffs have no ties.

### Filtering tree and interpretation




The climbing gym selects $F_1$: after the regret, EMV, and upside filters, its minimum profit of 71 thousand USD exceeds $F_6$'s 45.

## 5 Art Materials Orders [C9-2]

An art materials distributor selects an order plan $H_1$–$H_6$. Net profits are in hundred USD. The decision statistics are summarized below:

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $H_1$ | 142 | 88 | 81 | 122.2 |
| $H_2$ | 160 | 70 | 63 | 136.6 |
| $H_3$ | 196 | 34 | 54 | 123.1 |
| $H_4$ | 124 | 79 | 99 | 106 |
| $H_5$ | 223 | 7 | 81 | 116.8 |
| $H_6$ | 178 | 52 | 45 | 130.3 |

Apply these stages in order:

1. Maximin: Best 4.
2. Minimax Regret: Best 3.
3. Maximax: Best 2.
4. EMV/Bayes: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

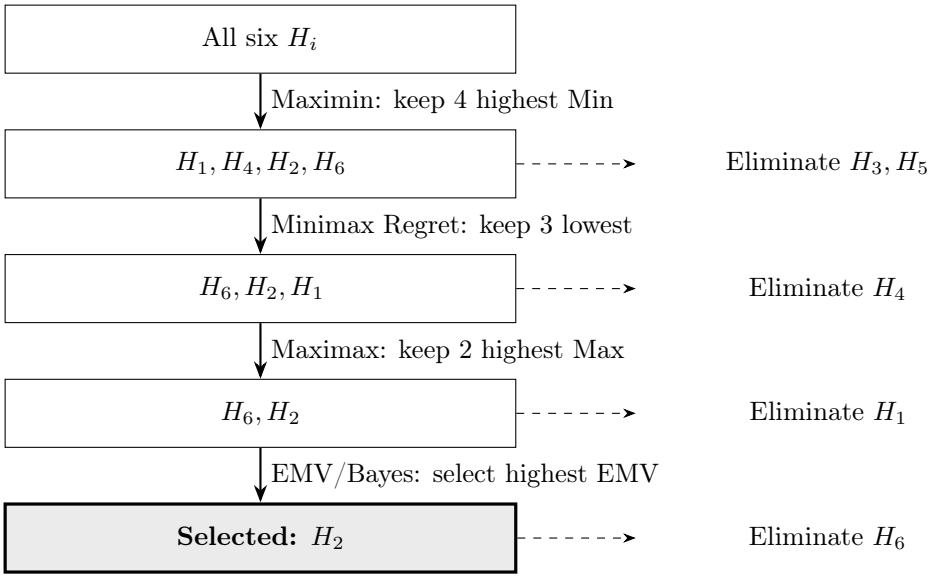
### Supporting filtering table

| Step | Rule | Survivors (relevant values) | Eliminated (values) |
|---|---|---|---|
| 0 | Initial alternatives | $H_5, H_3, H_6, H_2, H_1, H_4$ | — |
| 1 | Maximin: keep 4 highest Min | $H_1(88)$, $H_4(79)$, $H_2(70)$, $H_6(52)$ | $H_3(34)$, $H_5(7)$ |
| 2 | Minimax Regret: keep 3 lowest | $H_6(45)$, $H_2(63)$, $H_1(81)$ | $H_4(99)$ |
| 3 | Maximax: keep 2 highest Max | $H_6(178)$, $H_2(160)$ | $H_1(142)$ |
| 4 | EMV/Bayes: select highest EMV | **Selected:** $H_2(136.6)$ | $H_6(130.3)$ |

The cutoffs are Min 52, maximum regret 81, and Max 160; the final EMV comparison is 136.6 versus 130.3. There are no cutoff or final ties.



**Filtering tree and interpretation**




The art materials distributor selects $H_2$: after the downside, regret, and upside filters, its EMV of 136.6 hundred USD exceeds $H_6$'s 130.3.

## C10: Separate activity reminder (unscored)

Complete the individual **Interactive Decision Webpage: C10 Practice** activity and submit the deployed webpage URL. See the separate activity for instructions.

C10 example solution.

## C5 catch Up

**Camera Rental Plans (Question 1).** Apply Minimax Regret to the original payoff table.

$A_1$ Essential, $A_2$ Shared, $A_3$ Extensive, $A_4$ Modular.
$S_1$ High, $S_2$ Normal, $S_3$ Low.
Profits and regrets are in thousands of USD.
State bests are $\boldsymbol{b} = (235, 148, 62)$.
Regret is $r_{ij} = b_j - P_{ay}(A_i, S_j)$.

| Regret | $S_1$ | $S_2$ | $S_3$ | Max. |
|---|---|---|---|---|
| $A_1$ | $235 - 95 = 140$ | $148 - 88 = 60$ | $62 - 62 = 0$ | 140 |
| $A_2$ | $235 - 135 = 100$ | $148 - 148 = 0$ | $62 - 32 = 30$ | 100 |
| $A_3$ | $235 - 235 = 0$ | $148 - 46 = 102$ | $62 - (-38) = 100$ | 102 |
| $A_4$ | $235 - 165 = 70$ | $148 - 125 = 23$ | $62 - (-18) = 80$ | 80 |

$$\min\{140, 100, 102, 80\} = 80 \implies A_4 \text{ (Modular)} \quad \text{Maximum regret } 80$$

# 6 Summarized Solutions [C8–C9]

## 6.1 Camera Rental Plans C8

Values in thousands of USD.

**(a) Original payoffs**

| Criterion | Choice | Value |
|---|---|---|
| Maximax | $A_3$ | 235 |
| Maximin | $A_1$ | 62 |
| Minimax Regret | $A_4$ | 80 |
| EMV/Bayes | $A_2$ | 109.3 |

**(b) Eligibility**

| Alternative | Eligible range of $s$ |
|---|---|
| $A_1$ | $0 \le s \le 90$ |
| $A_2$ | $0 \le s \le 90$ |
| $A_3$ | $38 \le s \le 90$ |
| $A_4$ | $18 \le s \le 90$ |

**(c) Maximin**

- Compare eligible minima $m_1 = 62$ and $m_4 = s - 18$. Both $m_2 = 32$ and $m_3 \le 46$ are below $m_1$.
- $A_1$ and $A_4$ tie at $s = 80$ with minimum 62.
- First sole win for $A_4$ at $s = 81$, with minimum 63.

## 6.2 Outdoor Cinema Refreshments C8

Values in hundreds of USD.

**(a) Original payoffs**

| Criterion | Choice | Value |
|---|---|---|
| Maximax | $C_3$ | 176 |
| Maximin | $C_1$ | 60 |
| Minimax Regret | $C_4$ | 40 |
| EMV/Bayes | $C_4$ | 91 |

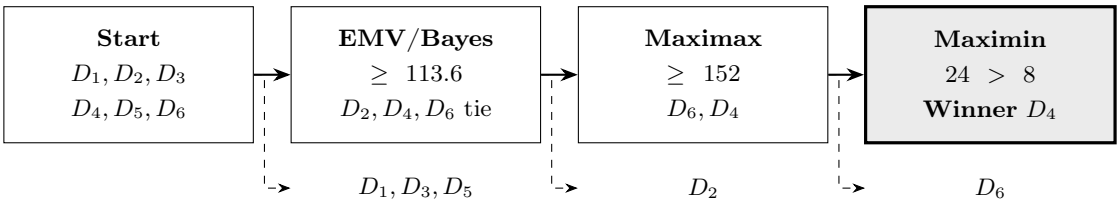
**(b) Eligibility**

| Alternative | Eligible range of $g$ |
|---|---|
| $C_1$ | $92 \le g \le 156$ |
| $C_2$ | None |
| $C_3$ | None |
| $C_4$ | $92 \le g \le 156$ |

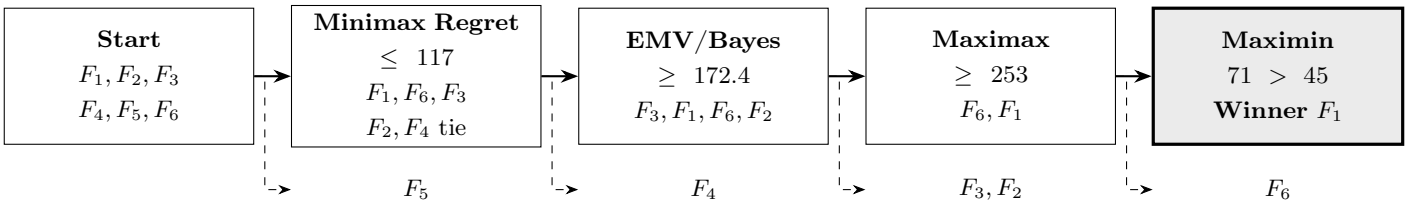
**(c) Minimax Regret**

- Eligible state bests (high, low) from $C_1, C_4$ are $(\max(g, 136), 60)$.
- Maximum regrets are $R_1 = \max(136 - g, 0)$ and $R_4 = \max(g - 136, 14)$.
- $C_1$ and $C_4$ tie at $g = 122$ with regret 14.
- First sole win for $C_1$ at $g = 123$, where $13 < 14$.

## 6.3 Garden Centre Stock C9



## 6.4 Climbing Gym Expansion C9




## 6.5 Art Materials Orders C9

